

NCERT Solutions Class 12 Computer Science Chapter 9 Structured Query Language (SQL)

1 (a) Define RDBMS. Name any two RDBMS software.

Answer

An **RDBMS** is a database system that follows the relational model. It organises information in table structures (rows & columns) and lets you relate, query, and modify that data using SQL. Common examples include **MySQL** and **Oracle Database**.

1 (b) What is the purpose of the following clauses in a select statement?

(i) ORDER BY

Answer: Arranges the resulting rows by one or more columns. By default it's ascending; use DESC for descending and ASC to be explicit about ascending.

(ii) GROUP BY

Answer: Collects rows into groups that share the same values in specified column(s). It's typically paired with aggregate functions such as SUM, COUNT, or AVG to produce summaries per group.

1(c) Site any two differences between Single Row Functions and Aggregate Functions.

Answer

Single Row Functions	Aggregate Functions
Evaluate one row at a time and yield one value for each row.	Process a set of rows and return a single value per set or per group.
Usable in SELECT, WHERE, ORDER BY, etc.	Used in SELECT/HAVING, often alongside GROUP BY.

1(d) What do you understand by Cartesian Product?

Answer

A **Cartesian Product** pairs every row of one table with every row of another, producing all possible combinations. In SQL this corresponds to a CROSS JOIN (or listing tables without a join condition).

1(e) Differentiate between the following statements:

(i) ALTER and UPDATE

ALTER statement	UPDATE statement
Changes the <i>definition</i> of database objects (e.g., add/drop columns, keys, indexes).	Changes the <i>stored values</i> in existing rows/columns.
Example: ALTER TABLE Employees ADD Email VARCHAR(255);	Example: UPDATE Employees SET Email='john.doe@example.com' WHERE EmployeeID=101;

(ii) DELETE and DROP

DELETE statement	DROP statement
Removes selected rows; the table and its structure remain intact.	Removes the object itself (e.g., table/view/index) along with its data and structure.
DELETE FROM Employees WHERE Department='Marketing';	DROP TABLE Products;

1 (f) Write the name of the functions to perform the following operations:

Answer

Requirement	Function
Show the weekday for 15 Aug 1947	DAYNAME('1947-08-15')
Extract n characters from a position	SUBSTRING(str, pos, n)
Return month name of a given date	MONTHNAME('yyyy-mm-dd')
Convert a name to uppercase	UPPER('YourName')

2. Write the output produced by the following SQL statements:

(a) SELECT POW(2, 3);

POW(2, 3)
8

(b) SELECT ROUND(342.9234, -1);

ROUND(342.9234, -1)
340

(c) SELECT LENGTH("Informatics Practices");

LENGTH("Informatics Practices")
21

**(d) SELECT YEAR("1979/11/26"), MONTH("1979/11/26"),
DAY("1979/11/26"), MONTHNAME("1979/11/26");**

YEAR("1979/11/26")	MONTH("1979/11/26")	DAY("1979/11/26")	MONTHNAME("1979/11/26")
1979	11	26	November

(e)

**SELECT LEFT("INDIA", 3),
RIGHT("ComputerScience", 4),
MID("Informatics", 3, 4),
SUBSTR("Practices", 3);**

LEFT("INDIA", 3)	RIGHT("ComputerScience", 4)	MID("Informatics", 3, 4)	SUBSTR("Practices", 3)
IND	ence	form	actices

3. Consider the following MOVIE table and write the SQL queries based on it.

Mov ieID	MovieNa me	Cate gory	Release Date	Productio nCost	Business Cost
001	Hindi_M ovie	Musi cal	2018-04 -23	124500	130000
002	Tamil_M ovie	Actio n	2016-05 -17	112000	118000
003	English_ Movie	Horr or	2017-08 -06	245000	360000
004	Bengali_ Movie	Adve nture	2017-01 -04	72000	100000
005	Telugu_ Movie	Actio n	-	100000	-
006	Punjabi_ Movie	Com edy	-	30500	-

Answer

(a) Display all the information from the Movie table.

SELECT * FROM Movie;

MOVIEID	MOVIE NAME	CATEGORY	RELEASE DATE	PRODUCTION COST	BUSINESS COST
1	Hindi_Movie	Musical	2018-04-23	124500	130000
2	Tamil_Movie	Action	2016-05-17	112000	118000
3	English_Movie	Horror	2017-08-06	245000	360000
4	Bengali_Movie	Adventure	2017-01-04	72000	100000
5	Telugu_Movie	Action	NULL	100000	NULL
6	Punjabi_Movie	Comedy	NULL	30500	NULL

(b) List business done by the movies showing only MovieID, MovieName and Total_Earning.

```
SELECT MovieID, MovieName, (ProductionCost + BusinessCost) AS
Total_Earning
FROM Movie
WHERE ReleaseDate IS NOT NULL;
```

MovieID	MovieName	Total_Earning
1	Hindi_Movie	254500
2	Tamil_Movie	230000
3	English_Movie	605000
4	Bengali_Movie	172000

(c) List the different categories of movies.

```
SELECT DISTINCT Category FROM MOVIE;
```

Category
Musical
Action

Category
Horror
Adventure
Comedy

(d) Find the net profit of each released movie.

```
SELECT MovieID, MovieName, BusinessCost - ProductionCost AS
NetProfit
FROM Movie
WHERE ReleaseDate IS NOT NULL;
```

MovieID	MovieName	NetProfit
1	Hindi_Movie	5500
2	Tamil_Movie	6000
3	English_Movie	115000
4	Bengali_Movie	28000

(e) List MovieID, MovieName and Cost for movies with ProductionCost > 10,000 and < 1,00,000.

```
SELECT MovieID, MovieName, ProductionCost AS Cost
FROM MOVIE
WHERE ProductionCost > 10000 AND ProductionCost < 100000;
```

MovieID	MovieName	Cost
4	Bengali_Movie	72000
6	Punjabi_Movie	30500

(f) List details of all movies which fall in Comedy or Action.

```
SELECT * FROM MOVIE
WHERE Category = 'Comedy' OR Category = 'Action';
```

MOVIEID	MOVIE NAME	CATEGORY	RELEASE DATE	PRODUCTION COST	BUSINESS COST
2	Tamil_Movie	Action	2016-05-17	112000	118000
5	Telugu_Movie	Action	NULL	100000	NULL

MOVIEID	MOVIE NAME	CATEGORY	RELEASE DATE	PRODUCTION COST	BUSINESS COST
6	Punjabi_Movie	Comedy	NULL	30500	NULL

(g) List details of all movies which have not been released yet.

SELECT * FROM MOVIE
WHERE ReleaseDate IS NULL;

MOVIEID	MOVIE NAME	CATEGORY	RELEASE DATE	PRODUCTION COST	BUSINESS COST
5	Telugu_Movie	Action	NULL	100000	NULL
6	Punjabi_Movie	Comedy	NULL	30500	NULL

4. Suppose your school management has decided to conduct cricket matches between students of Class XI and Class XII. Students of each class are asked to join any one of the four teams – Team Titan, Team Rockers, Team Magnet and Team Hurricane. During summer vacations, various matches will be conducted between these teams. Help your sports teacher to do the following:

(a) Create a database "Sports".

(b) Create a table "TEAM" with following considerations:

1. It should have a column TeamID for storing an integer value between 1 to 9, which refers to unique identification of a team.
2. Each TeamID should have its associated name (TeamName), which should be a string of length not less than 10 characters.

(c) Using table level constraint, make TeamID as the primary key.

(d) Show the structure of the table TEAM using a SQL statement.

(e) As per the preferences of the students four teams were formed as given below. Insert these four rows in TEAM table:

Row 1: (1, Team Titan)

Row 2: (2, Team Rockers)

Row 3: (3, Team Magnet)

Row 4: (4, Team Hurricane)

(f) Show the contents of the table TEAM using a DML statement.

(g) Now create another table MATCH_DETAILS and insert data as shown below. Choose appropriate data types and constraints for each attribute.

Table: MATCH_DETAILS

MatchID	MatchDate	FirstTeamID	SecondTeamID	FirstTeamScore	SecondTeamScore
M1	2018-07-17	1	2	90	86
M2	2018-07-18	3	4	45	48
M3	2018-07-19	1	3	78	56
M4	2018-07-19	2	4	56	67
M5	2018-07-18	1	4	32	87

M6	2018-07 -17	2	3	67	51
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Answer:

(a) Create the database
CREATE DATABASE Sports;

(b) Create the TEAM table
CREATE TABLE TEAM (
TeamID INT,
TeamName VARCHAR(20)
);

(c) Declare the primary key
ALTER TABLE TEAM
ADD PRIMARY KEY (TeamID);

(d) Display the table structure
DESCRIBE TEAM;

Field	Type	Null	Key	Default	Extra
TeamID	int	NO	PRI	NULL	

Field	Type	Null	Key	Default	Extra
TeamName	char(20)	YES		NULL	

(e) Insert the four teams

INSERT INTO TEAM (TeamID, TeamName) VALUES

(1, 'Team Titan'),
(2, 'Team Rockers'),
(3, 'Team Magnet'),
(4, 'Team Hurricane');

(f) Show all rows in TEAM

SELECT * FROM TEAM;

TeamID	TeamName
1	Team Titan
2	Team Rockers
3	Team Magnet
4	Team Hurricane

(g) Create MATCH_DETAILS, insert matches, and view data

```
CREATE TABLE MATCH_DETAILS (  
    MatchID    VARCHAR(10),  
    MatchDate   DATE,  
    FirstTeamID INT,  
    SecondTeamID INT,  
    FirstTeamScore INT,  
    SecondTeamScore INT,  
    CONSTRAINT PK_MatchID PRIMARY KEY (MatchID)  
);
```

```
INSERT INTO MATCH_DETAILS  
(MatchID, MatchDate, FirstTeamID, SecondTeamID, FirstTeamScore,  
SecondTeamScore) VALUES  
( 'M1', '2018-07-17', 1, 2, 90, 86),  
( 'M2', '2018-07-18', 3, 4, 45, 48),  
( 'M3', '2018-07-19', 1, 3, 78, 56),  
( 'M4', '2018-07-19', 2, 4, 56, 67),  
( 'M5', '2018-07-18', 1, 4, 32, 87),  
( 'M6', '2018-07-17', 2, 3, 67, 51);
```

```
SELECT * FROM MATCH_DETAILS;
```

MatchID	MatchDate	FirstTeamID	SecondTeamID	FirstTeamScore	SecondTeamScore
M1	2018-07-17	1	2	90	86
M2	2018-07-18	3	4	45	48
M3	2018-07-19	1	3	78	56
M4	2018-07-19	2	4	56	67
M5	2018-07-18	1	4	32	87
M6	2018-07-17	2	3	67	51

5. Using the sports database ... write the queries for the following:

Answer

(a) Both teams scored more than 70.

```
SELECT MatchID
FROM MATCH_DETAILS
WHERE FirstTeamScore > 70 AND SecondTeamScore > 70;
```

Output:

```
+-----+
| MatchID |
+-----+
| M1      |
+-----+
```

(b) FirstTeam < 70 and SecondTeam > 70.

```
SELECT MatchID
FROM MATCH_DETAILS
WHERE FirstTeamScore < 70 AND SecondTeamScore > 70;
```

Output

```
+-----+
| MatchID |
+-----+
| M5      |
+-----+
```

(c) Matches played by Team 1 and won by it.

```
SELECT MatchID, MatchDate
FROM MATCH_DETAILS
WHERE FirstTeamID = 1 AND FirstTeamScore > SecondTeamScore;
```

Output

```
+-----+-----+
| MatchID | MatchDate |
+-----+-----+
```

```
+-----+-----+
| M1      | 2018-07-17 |
| M3      | 2018-07-19 |
+-----+-----+
```

(d) Matches played by Team 2 and not won by it.

```
SELECT MatchID
FROM MATCH_DETAILS
WHERE SecondTeamID = 2 AND SecondTeamScore <=
FirstTeamScore;
```

Output

```
+-----+
| MatchID |
+-----+
| M1      |
+-----+
```

(e) Rename relation and columns.

```
ALTER TABLE TEAM RENAME TO T_DATA;
ALTER TABLE T_DATA CHANGE COLUMN TeamID T_ID INT;
ALTER TABLE T_DATA CHANGE COLUMN TeamName T_NAME
CHAR(20);
```

6. A shop called Wonderful Garments who sells school uniforms maintains a database SCHOOLUNIFORM as shown below. It consisted of two relations - UNIFORM and COST. They made UniformCode as the primary key for UNIFORM relations. Further, they used UniformCode and Size to be composite keys for COST relation. By analysing the database

schema and database state, specify SQL queries to rectify the following anomalies.

(a) M/S Wonderful Garments also keeps handkerchiefs of red colour, medium size of Rs. 100 each.

(b) INSERT INTO COST (UCode, Size, Price) values (7, 'M', 100); When the above query is used to insert data, the values for the handkerchief without entering its details in the UNIFORM relation is entered. Make a provision so that the data can be entered in the COST table only if it is already there in the UNIFORM table.

(c) Further, they should be able to assign a new UCode to an item only if it has a valid UName. Write a query to add appropriate constraints to the SCHOOLUNIFORM database.

(d) Add the constraint so that the price of an item is always greater than zero.

Answer:

(b)

```
ALTER TABLE COST ADD CONSTRAINT fk_uniform_ucode_size FOREIGN KEY  
(UCode) REFERENCES UNIFORM (UCode);
```

(c)

```
ALTER TABLE UNIFORM ADD CONSTRAINT CK_UName_UCode
CHECK (UName IS NOT NULL);
```

(d)

```
ALTER TABLE COST ADD CONSTRAINT CK_Price_Positive CHECK (Price > 0);
```

7. Consider the following table named "Product", showing details of products being sold in a grocery shop:

PCode	PName	UPrice	Manufacturer
P01	Washing Powder	120	Surf
P02	Toothpaste	54	Colgate
P03	Soap	25	Lux
P04	Toothpaste	65	Pepsodent
P05	Soap	38	Dove
P06	Shampoo	245	Dove

Write SQL queries for the following:

(a) Create the table Product with appropriate data types and constraints.

(b) Identify the primary key in Product.

(c) List the Product Code, Product name and price in descending order of their product name. If PName is the same, then display the data in ascending order of price.

(d) Add a new column Discount to the table Product.

(e) Calculate the value of the discount in the table Product as 10 per cent of the UPrice for all those products where the UPrice is more than 100, otherwise the discount will be 0.

(f) Increase the price by 12 per cent for all the products manufactured by Dove.

(g) Display the total number of products manufactured by each manufacturer.

Write the output(s) produced by executing the following queries on the basis of the information given above in the table Product:

(h) SELECT PName, avg(UPrice) FROM Product GROUP BY Pname;

- (i) SELECT DISTINCT Manufacturer FROM Product;
- (j) SELECT COUNT (DISTINCT PName) FROM Product;
- (k) SELECT PName, MAX(UPrice), MIN(UPrice) FROM Product GROUP BY PName;

Answer:

(a) Create the table Product.

```
CREATE TABLE Product (  
    PCode    VARCHAR(10) PRIMARY KEY,  
    PName    VARCHAR(50) NOT NULL,  
    UPrice   INT NOT NULL CHECK (UPrice >= 0),  
    Manufacturer VARCHAR(50) NOT NULL  
);
```

(b) Identify the primary key in Product.

Answer: PCode is the primary key.

(c) List PCode, PName, UPrice ordered by PName (DESC), then UPrice (ASC).

```
SELECT PCode, PName, UPrice  
FROM Product  
ORDER BY PName DESC, UPrice ASC;
```

Output:

PCode	PName	UPrice
P01	WASHING POWDER	120
P02	TOOTHPASTE	54
P04	TOOTHPASTE	65
P03	SOAP	25
P05	SOAP	38
P06	SHAMPOO	245

(d) Add a new column Discount.

ALTER TABLE Product ADD COLUMN Discount FLOAT;

(e) Set Discount = 10% when UPrice > 100, else 0.

UPDATE Product

SET Discount = CASE WHEN UPrice > 100 THEN UPrice * 0.10 ELSE 0 END;

Output:

PCode	PName	UPrice	Manufacturer	Discount
P01	WASHING POWDER	120	SURF	12
P02	TOOTHPASTE	54	COLGATE	0
P03	SOAP	25	LUX	0

P04	TOOTHPASTE	65	PEPSODENT	0
P05	SOAP	38	DOVE	0
P06	SHAMPOO	245	DOVE	24.5

(f) Raise price by 12% for products by Dove.

UPDATE Product

SET UPrice = ROUND(UPrice * 1.12)

WHERE Manufacturer = 'Dove';

PCode	PName	UPrice	Manufacturer	Discount
P01	WASHING POWDER	120	SURF	12
P02	TOOTHPASTE	54	COLGATE	0
P03	SOAP	25	LUX	0
P04	TOOTHPASTE	65	PEPSODENT	0
P05	SOAP	43	DOVE	0
P06	SHAMPOO	274	DOVE	24.5

(g) Count products by each manufacturer.

SELECT Manufacturer, COUNT(*) AS TotalProducts

FROM Product

GROUP BY Manufacturer;

Manufacturer	TotalProducts
SURF	1
COLGATE	1
LUX	1
PEPSODENT	1
DOVE	2

(h) SELECT PName, avg(UPrice) FROM Product GROUP BY Pname;

PName	avg(UPrice)
WASHING POWDER	120.0000
TOOTHPASTE	59.5000
SOAP	34.0000
SHAMPOO	274.0000

(i) SELECT DISTINCT Manufacturer FROM Product;

Manufacturer
SURF
COLGATE
LUX
PEPSODENT
DOVE

(j) SELECT COUNT (DISTINCT PName) FROM Product;

COUNT(DISTINCT PName)
4

(k) SELECT PName, MAX(UPrice), MIN(UPrice) FROM Product GROUP BY PName;

PName	MAX(UPrice)	MIN(UPrice)
WASHING POWDER	120	120
TOOTHPASTE	65	54

PName	MAX(UPrice)	MIN(UPrice)
SOAP	43	25
SHAMPOO	274	274

8. Using the CARSHOWROOM database given in the chapter, write the SQL queries for the following:

(a) Add a new column Discount in the INVENTORY table.

(b) Set appropriate discount values for all cars keeping in mind the following:

- 1. No discount is available on the LXI model.**
- 2. VXI model gives a 10 per cent discount.**
- 3. A 12 per cent discount is given on cars other than LXI model and VXI model.**

(c) Display the name of the costliest car with fuel type "Petrol".

(d) Calculate the average discount and total discount available on Baleno cars.

(e) List the total number of cars having no discount.

Answer

Table inventory:

CarId	CarName	Price	Model	YearManufacture	FuelType	FinalPrice
D001	Dzire	582613.00	LXI	2017	Petrol	652526.6
D002	Dzire	673112.00	VXI	2018	Petrol	753885.4
B001	Baleno	567031.00	Sigma1.2	2019	Petrol	635074.7
B002	Baleno	647858.00	Delta1.2	2018	Petrol	725601.0
E001	EECO	355205.00	5 STR STD	2017	CNG	397829.6
E002	EECO	654914.00	CARE	2018	CNG	733503.7
S001	SWIFT	514000.00	LXI	2017	Petrol	575680.0
S002	SWIFT	614000.00	VXI	2018	Petrol	687680.0

(a) Add a new column Discount in INVENTORY.

ALTER TABLE INVENTORY

ADD COLUMN Discount FLOAT;

(b) Set the discounts per model.

UPDATE INVENTORY SET Discount = 0 WHERE Model = 'LXI';

UPDATE INVENTORY SET Discount = Price * 0.10 WHERE Model = 'VXI';

UPDATE INVENTORY

SET Discount = Price * 0.12
WHERE Model NOT IN ('LXI','VXI');

Output:

CarId	CarName	Price	Model	YearManufacture	FuelType	FinalPrice	Discount
D001	Dzire	582613.00	LXI	2017	Petrol	652526.60	0
D002	Dzire	673112.00	VXI	2018	Petrol	753885.40	67311.2
B001	Baleno	567031.00	Sigma1.2	2019	Petrol	635074.70	68043.7
B002	Baleno	647858.00	Delta1.2	2018	Petrol	725601.00	77743
E001	EECO	355205.00	5 STR STD	2017	CNG	397829.60	42624.6
E002	EECO	654914.00	CARE	2018	CNG	733503.70	78589.7
S001	SWIFT	514000.00	LXI	2017	Petrol	575680.00	0
S002	SWIFT	614000.00	VXI	2018	Petrol	687680.00	61400

(c) Name of the costliest Petrol car.

```
SELECT CarName
FROM INVENTORY
WHERE FuelType = 'Petrol'
      AND Price = (SELECT MAX(Price) FROM INVENTORY WHERE
FuelType='Petrol');
```

Output:

```
+-----+
| CarName |
```

```
+-----+
| Dzire |
+-----+
```

(d) Average and total discount on Baleno cars.

```
SELECT AVG(Discount) AS AverageDiscount,
       SUM(Discount) AS TotalDiscount
FROM INVENTORY
WHERE CarName='Baleno';
```

Output:

```
+-----+-----+
| AverageDiscount | TotalDiscount |
+-----+-----+
| 72893.33984375 | 145786.6796875 |
+-----+-----+
```

(e) Count cars with no discount.

```
SELECT COUNT(*)
FROM INVENTORY
WHERE Discount = 0;
```

Output:

```
+-----+
| COUNT(*) |
+-----+
```

| 2 |

+-----+

All questions are unchanged; only answer wording has been refreshed to ensure originality while preserving meaning and correctness.

